



Globally networked: Intraorganizational boundary spanning in the global organization

Torben Pedersen^{a,1}, Giuseppe Soda^{a,c,1}, Diego Stea^{b,*,1}

^a Bocconi University, Italy

^b Copenhagen Business School, Denmark

^c SDA Bocconi School of Management, Italy

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ABSTRACT

Multinational corporations (MNCs) need to sense, source, and mobilize knowledge when and where it arises, whether at home, or elsewhere in the world. For this reason, MNCs benefit from employee networks of relationships that span across intraorganizational barriers, allowing for the efficient mobilization of knowledge across boundaries. Yet, which organizational members are more likely to be able to develop these boundary spanning networks? We leverage a unique data set from a large multinational corporation to empirically test a comprehensive model that captures the effect of an employee's mandate, expertise, and behavioral orientations on her likelihood to span intraorganizational boundaries that manifest themselves in the form of hierarchies, intra-functional domains, and geographic territories. We find that the employees that are more likely to be boundary spanners are those having mandates with a global impact, high levels of expertise, and a collaborative orientation in their networking behaviors. In addition, we find that these effects are stronger for those employees that have large formal workflow networks.

1. Introduction

In the International Business (IB) literature it has been highlighted that MNCs' competitive *raison d'être* is cross-border knowledge creation, as well as transfer, and exploitation (Doz, Santos, & Williamson, 2001; Kogut & Zander, 1993). As the pace of technological change has accelerated, MNCs no longer have the luxury of simply cultivating home-grown responses to innovations by rivals. Staying abreast of requirements for continuous developments and innovations entails MNCs to sense, source, and mobilize new knowledge streams when and where they arise, whether at home or elsewhere in the world. Recognizing this aspect, IB scholars devoted considerable attention to the study of intraorganizational knowledge mobilization (Gupta & Govindarajan, 2000; Minbaeva, Pedersen, Björkman, Fey, & Park, 2003; Monteiro, Arvidsson, & Birkinshaw, 2008).

While most IB studies have focused on the organizational level, more recently, several scholars have argued for a focus on people, stressing the importance of exploring the micro-foundations of knowledge mobilization between the employees of MNCs (Foss, Husted, & Michailova, 2010; Foss & Pedersen, 2004; Haas & Cummings, 2014). The intuition behind this perspective is that it is not organizations, but

the individual employees that tap into, and make use of knowledge for the benefit of organizations. Thus, individual employees are the key players building those informal and emergent networks that lubricate and complement the formal organizational processes and structures designed to guide decision making and knowledge mobilization (McEvily, Soda, & Tortoriello, 2014).

MNCs bring together individuals with very different profiles, from distant parts of the world, and of the organization (Earley & Gibson, 2002), thus offering a good platform for the formation of boundary spanning relations (Schotter, Mudambi, Doz, & Gaur, 2017). Previous research has devoted particular attention to three types of boundaries arising from the activities carried out in MNCs. Geographical boundaries, such as working in different countries, world regions, or international time zones (Cramton, 2001; Espinosa, Cummings, Wilson, & Pearce, 2003; Hinds & Mortensen, 2005). Structural boundaries, such as working in different functions or functional areas, divisions, or business units (Bartlett, Ghoshal, & Birkinshaw, 2003; Dasí, Pedersen, Gooderham, Elter, & Hildrum, 2017; Puranam, Raveendran, & Knudsen, 2012). Hierarchical boundaries, arising from differences in tenure, or formal levels in the organizational hierarchy (Reagans, Zuckerman, & McEvily, 2004; Zenger & Lawrence, 1989).

* Corresponding author.

E-mail addresses: torben.pedersen@unibocconi.it (T. Pedersen), giuseppe.soda@unibocconi.it (G. Soda), ds.si@cbs.dk (D. Stea).

¹ Authors contributed equally.

Relationships that cut across these boundaries span multiple knowledge domains, give people the ability to absorb and convey complex ideas to diverse audiences (Reagans & McEvily, 2003), and in turn promote worldwide learning and overall performance (Doz et al., 2001; Kogut & Zander, 1993). Boundary spanners are therefore those employees who act as knowledge intermediaries between different parties within and outside their organizations (Birkinshaw, Ambos, & Bouquet, 2017; Monteiro & Birkinshaw, 2017). Boundary spanners play a key role in global organizations, particularly in terms of accessing and mobilizing knowledge resources (Bouquet & Birkinshaw, 2008; Schotter & Beamish, 2011). Because of these reasons, MNCs have strong interests in the development of boundary spanning ties within the organization (Caimo & Lomi, 2015; Haas & Cummings, 2014).

Yet, while prior studies have highlighted the importance of boundary spanning ties (Birkinshaw et al., 2017; Minbaeva & Santangelo, 2017; Monteiro & Birkinshaw, 2017; Schotter et al., 2017), research has also shown that the very diversity that makes those ties valuable, also makes them difficult to establish and maintain (Hargadon & Sutton, 1997; Tsai, 2001). Moreover, as highlighted in Schotter et al. (2017, p. 404): “little is known about the characteristics of boundary spanners in global organizations and whether their capabilities are inherent or can be developed”. Taken together, this raises the important question: Which employees are more likely to develop work-related networks that crosscut intraorganizational barriers?

In order to answer this question, we leverage a unique data set from a large multinational corporation to empirically test a comprehensive model that captures the effect of an employee's mandate, expertise, and behavioral orientations on her likelihood to span intraorganizational boundaries that manifest themselves in the form of hierarchies, intra-functional domains, and geographic territories. Based on opportunity-capacity-willingness theories of behavior (Blumberg & Pringle, 1982), we propose that the employees with work mandates that the organization regards as having a global impact, high levels of expertise, and collaborative networking attitudes should be more likely to have boundary spanning ties in their networks. We also propose that these mechanisms should be sensitive to the size of an employee's formal workflow network. Our analysis of network data from a survey of 275 employees supports these arguments.

This paper makes a number of contributions. First, we leverage recent discussions highlighting a lack of knowledge about how boundary spanning works (Birkinshaw et al., 2017; Monteiro & Birkinshaw, 2017), and more generally about the factors that affect the effectiveness of boundary spanning (Schotter et al., 2017). As highlighted by Schotter et al. (2017, p. 413): “individual actors play an important role in the effectiveness of boundary spanning. [...] However, not every individual may be equally effective in the boundary spanning function”. Our first contribution is to advance the discussion aimed at better characterizing the boundary spanners of global organizations by showing that the employees that are more likely to develop work-related networks that crosscut intraorganizational barriers are those with the right mandate, expertise, and behavioral attitudes.

Second, boundaries in global organizations may be of different nature. Most traditionally, IB research has distinguished between geographic, functional, and hierarchical boundaries. Very often, however, research on boundary spanning has focused on one specific type of boundaries at a time (Birkinshaw et al., 2017; Minbaeva & Santangelo, 2017; Monteiro & Birkinshaw, 2017; Roberts & Beamish, 2017). Different from this approach, our model puts forward the idea that many of the factors that can be associated with boundary spanning activities can simultaneously facilitate the spanning of boundaries that manifest themselves in the form of hierarchies, intra-functional domains, and geographic territories. Our results confirm these intuitions, showing that boundary spanning simultaneously happens on all the three dimensions. Yet, to further investigate the nature of these results, in post hoc analyses we also disaggregated our dependent variable into the three key dimensions of geographical, intra-functional and hierarchical

boundary spanning, and explored the specific impact of our explanatory variables on each dimension separately. By doing so, we provide initial evidence on which aspects matter the most for the spanning of specific boundaries, while controlling out for potential confounding factors.

Third, we relate to the criticism that has been raised against the more structuralist accounts of boundary spanning, which tend to conflate the “position” occupied and the “role” performed by the boundary spanners (Birkinshaw et al., 2017; Monteiro & Birkinshaw, 2017; Schotter et al., 2017; Soda, Stea, & Pedersen, 2019; Stea & Pedersen, 2017). Our contribution in this sense, is to put forward a model that simultaneously considers both the more structural drivers of boundary spanning (that is, the mandate of those employees that are formally assigned to roles with global implications, and the size of their formal workflow networks), but also the more behavioral aspects of how skilled employees are, and of how they engage in networking behaviors. Our final contribution is more empirical in nature. As recently pointed out by Schotter et al.: “existing research on boundary spanning and boundary spanners is predominantly conceptual or based on a limited number of case studies” (Schotter et al., 2017, p. 404). While building on many of the important theoretical insights developed in conceptual or case-based research, we are able to put our theoretical arguments to an empirical test using a rich primary dataset including 275 complete responses from the supply chain function of a large, global organization. In doing so, we join recent efforts (Minbaeva & Santangelo, 2017) aimed at answering the call to approach the study of boundary spanning also using quantitative approaches.

2. Theory and hypotheses

2.1. Boundary spanning networks in global organizations

A growing research stream conceptualizes the organizational context as one of network relations where employees are represented by nodes connected to one another via relational ties (Borgatti & Halgin, 2011; Brass, Galaskiewicz, Greve, & Tsai, 2004). In this view, ties constitute the informal emerging pipes through which information and knowledge is mobilized within the organization (Soda & Zaheer, 2012). Intraorganizational ties are thus important as much as they are unavoidable because they provide the behind-the-chart social infrastructure (Krackhardt & Hanson, 1993) that allows the mobilization of those knowledge resources that are necessary for organizational performance and survival (Borgatti & Foster, 2003; Wasserman & Faust, 1994). As MNCs' competitive *raison d'être* is cross-border knowledge mobilization, exploitation, and creation, MNCs have a strong interest that their organizational members establish networks of social relationships that allow them to mobilize knowledge readily with each other (Doz et al., 2001; Kogut & Zander, 1993).

In particular, due to their global nature, MNCs have the opportunity as well as necessity to mobilize knowledge across intraorganizational boundaries of different types. In particular, research has highlighted geographic boundaries, such as boundaries between different countries, or world regions (Cramton, 2001; Espinosa et al., 2003; Hinds & Mortensen, 2005); inter-functional boundaries (across the horizontal dimension of organizational structure), such as boundaries between individuals that work in different teams, functions, or business units (Bartlett et al., 2003; Dasí et al., 2017; Puranam et al., 2012); and hierarchical boundaries (across the vertical dimension of organizational structure), arising from the differences in the levels of the organizational hierarchy (Reagans et al., 2004; Zenger & Lawrence, 1989).

Based on these intuitions, and in line with the boundary spanning literature (Birkinshaw et al., 2017; Minbaeva & Santangelo, 2017; Schotter et al., 2017), we define an intraorganizational boundary spanning tie as a tie that cuts across geographical, intra-functional, or hierarchical boundaries. Thus, while concepts such as structural holes (Burt, 1992) or density (Coleman, 1988) refer to *structural properties* of social networks, the boundary spanning construct captures the *diversity*

in each employee's ego-network by measuring the heterogeneity of the colleagues (alters) linked to the focal employee (ego). More specifically, our dependent variable captures the composition of MNC employees' networks in terms of the geographical, and organizational (both horizontal and vertical) diversity that they embody.

The successful mobilization of knowledge across boundaries and distances of different types has been shown to be systematically related to desirable organizational outcomes. For example, connections to diverse knowledge pools ease knowledge mobilization within organizations (Reagans & McEvily, 2003), which in turn fosters creativity and innovation (Hennessey & Amabile, 2010). More broadly, and as eloquently put by Cohen and Levinthal (1990, p. 133), "interactions across individuals who each possess diverse and different knowledge structures will augment the organization's capacity of making novel linkages and associations—innovating—beyond what any individuals can achieve." Additionally, informal knowledge networks have a fundamental role in lubricating and complementing the workings of formal organizational processes and structures (McEvily et al., 2014), and this is particularly valuable where formal coordination and integration mechanisms need to be applied on a global scale, and are thus maximally complex and costly (Soda & Zaheer, 2012). In other words, informal networks of boundary spanning ties are extremely valuable in complementing formal integration mechanisms (Birkinshaw et al., 2017; McEvily et al., 2014; Minbaeva & Santangelo, 2017; Schotter et al., 2017).

However, besides being very valuable for organizations, ties that crosscut different intraorganizational barriers are also difficult to establish and sustain for a number of reasons (Caimo & Lomi, 2015). First, classic homophily considerations have led to the conclusion that similarity breeds connection (McPherson, Smith-Lovin, & Cook, 2001), thus relationships are more likely to arise within than across functional, geographical, as well as hierarchical boundaries. Second, a long research tradition shows that while individuals embedded in cohesive network structures are more likely to collaborate with one another (Coleman, 1988; Obstfeld, 2005), ties that link distant and distinct social circles tend to be associated with more opportunistic, and competitive behaviors (Xiao & Tsui, 2007). Thus, within-group relationships are less risky, and easier to manage than across-group relationships (Coleman, 1988). Third, formal design practices based on the conventional idea of pooling strongest interdependencies within purpose-built groups (Thompson, 1967) often end up crystallizing intraorganizational boundaries, and complicating communication patterns that span across boundaries (Tortoriello & Krackhardt, 2010). Fourth, as pointed out by Tushman and Scanlan (1981), since there are high costs for uncertainty, ambiguity, and difficulty in boundary spanning communications, individuals tend not to engage in boundary spanning relationships unless they perceive that doing so will be conducive to the mobilization of valuable knowledge otherwise difficult to obtain. Lastly, knowledge is harder to mobilize and integrate when the parties involved lack a common knowledge language (Reagans & McEvily, 2003; Tortoriello, Reagans, & McEvily, 2012). In sum, individuals weight the value of obtaining knowledge and the costs of contacting other individuals in their decision to create a tie (Nebus, 2006) and, as a result, organizational members may experience considerable difficulties in mobilizing, understanding, and absorbing knowledge generated across boundaries (Cohen & Levinthal, 1990; Stea & Pedersen, 2017).

If boundary spanning ties within the MNC are so important and valuable, and yet also so difficult to master and maintain, a key question arises: Which employees are more likely to develop work-related networks that crosscut intraorganizational barriers?

2.2. Opportunity, capacity, and willingness to mobilize knowledge across intraorganizational boundaries

To answer these questions, we provide a theoretical framework based on opportunity-capacity-willingness theories of behavior

(Blumberg & Pringle, 1982; Roberts & Beamish, 2017). In its broader terms, the framework proposes that the extent to which organizational actors are able to perform a given task depends on three dimensions. First, the opportunity to perform—as given by, for example, the presence of tools, working conditions, policies, procedures, and conditions that make the task possible. Second, the capability to perform—that is, whether the employee is sufficiently skilled to complete the task. Third, the intention and motivation to act. Consistent with the opportunity-capacity-willingness framework, our argument is that the extent to which employees have boundary spanning ties depends on whether they have assignments which the organization regards as having a global impact (i.e., opportunity), high levels of expertise (i.e., capacity), and a collaborative attitude in their networking orientations (i.e., willingness).

While social relationships originate from individuals' intention and motivation to connect with each other, organizational factors are fundamental enablers of the formation and persistence of ties. As Burt eloquently put it: "many relationships originate from factors, exogenous to the two people involved, that define opportunities for relations to form" (2000, p. 2). Thus, factors that bring employees together such as workspace location, procedures, structures, team and project assignments, or coordination and integration mechanisms are key for informal ties to arise.

Based on this intuition, our first contention is that if an employee has an assignment that the organization regards as one having a global impact for the organization, that should make it easier for the employee to establish ties with other organizational members who work across geographical, horizontal, and vertical boundaries of the organization. For example, in the context of the company where we conducted our study, Mrs. Emily F. is the Industrial Planner for the plant located at Dong Guan (China). She is formally part of the team "Operation China", thus reporting to the China Head of Planning (the company has several plants in China) whom is a direct report of the Supply Chain Vice President. However, because the Dong Guan plant is considered a "global" manufacturing platform available for all commercial operations, the scope of her job extends outside the region Asia. Mandates with global impact may include different responsibilities, such as conducting projects that affect practices at the global level, coordinating processes which are designed to cross geographies such as global production planning, as well as working for services which are centrally designed, and yet delivered at a global level for all the geographies, such as investment planning.

As the example illustrates, employees with mandates with a global impact do not necessarily need to operate across geographies, functions, or levels, but their assignments have global implications. Because of this, we propose that these assignments should be more likely to be embedded into organizational processes which tend to be more critical, and also more complex. As a result of this criticality and complexity, tasks with a global impact should require higher levels of integration, not only across different geographic locations, but also across intra-functional, and hierarchical boundaries. For example, the Logistic Networks function of the company where we conducted our study is in charge of designing activities across manufacturing, warehouses, and distribution centers which are delivered on a global scale, requiring a careful understanding of how these activities fit into the different regions of the global landscape. Such an understanding is likely developed by means of broad information gathering at different intra-functional levels (e.g., assessing the fit of the new services all along the value chain), as well as different hierarchical levels (e.g., discussing the strategic fit of the new services with top management, but also concrete implementation procedures with middle management). In sum, the greater complexity of those jobs that have implications that apply on a global scale triggers the need for flexible and less formalized information exchange with people operating across different boundaries of the organization. Informal networks of relationships are more dynamic and adaptable than formal channels, and may provide to individuals with

global work mandates the social infrastructure that allows them to interact and coordinate more efficiently and effectively. Opportunities as well as incentives to develop similar boundary spanning relationships should be more limited for employees whose mandates do not have a global impact.

Hypothesis 1. *Employees with mandates with a global impact are more likely than employees with mandates that do not have a global impact to have boundary spanning networks.*

While social relationships constitute the channels through which valuable knowledge and information is mobilized, they also entail costs (Borgatti & Halgin, 2011). Besides the natural effort and commitment involved in establishing, and maintaining any given work relationship, research shows that ties that mobilize non-redundant information might be particularly difficult to handle (Stea & Pedersen, 2017; Zhou, Shin, Brass, Choi, & Zhang, 2009). The processing of new knowledge depends in part on the related knowledge acquired in the past (Cohen & Levinthal, 1990; Obstfeld, 2005). Hence, ties that are conducive to the mobilization of non-redundant knowledge are more difficult to establish and maintain than those that carry redundant knowledge (Reagans & McEvily, 2003). This should be particularly relevant for ties that cut across geographical, hierarchical, and intra-functional boundaries. These are in fact ties that link highly different colleagues, and are thus likely to mobilize non-redundant knowledge, which makes them extremely valuable, and yet also potentially difficult to establish and maintain.

The social environment categorizes individuals as a function of their expertise, creating an expectation of the value that could be derived from activating a social connection (Shipilov, Gulati, Kilduff, Li, & Tsai, 2014). Individuals who are highly knowledgeable are perceived as capable of satisfying demands from diverse constituencies (Shipilov & Li, 2012; Shipilov et al., 2014), in addition to that, experts are more likely to be assigned complex tasks, which in turn should make them more likely to interact with different types of people across the organization. Furthermore, high expertise allows the mobilization of multiple forms of knowledge which enable the possibility to satisfy the requests from diverse partners fostering the formation of ties with different groups (Beckman, Schoonhoven, Rottner, & Kim, 2014). Based on these intuitions, we argue that employees who are regarded as experts in their fields will be more likely to have boundary spanning networks than employees that are not experts in their fields.

Especially in knowledge intensive settings, the need for knowledge resources is a fundamental driver of networking action (Cannella & McFadyen, 2016). This view is consistent with the overall idea that ties are used by social actors as pipes for the mobilization of knowledge resources (Borgatti & Halgin, 2011). Thus, employees in knowledge intensive settings should be particularly interested in being linked to knowledgeable colleagues. Specifically, employees have higher expectations of the potential returns of interacting with experts, and this should motivate them to establish ties with expert colleagues. In addition to this, and precisely because of their competence and skills, experts should also be more visible across geographically or functionally distant parts of the organization. Thus, expert employees should be more likely than non-expert employees to be both known, and sought after by colleagues located across geographical, intra-functional, or hierarchical boundaries.

Additionally, high expertise constitutes the intellectual capital that allows employees to communicate their knowledge to, and understand the knowledge possessed by diverse others (Reinholt, Pedersen, & Foss, 2011). Expert employees are those that possess extensive knowledge. Having extensive prior knowledge increases the probability that incoming new knowledge is related to what is already known, and thus provides the basis for absorbing new knowledge (Cohen & Levinthal, 1990). On the other hand, extensive knowledge also makes employees better equipped to understand when and how their own knowledge may be valuable for colleagues who work in different contexts, facilitating

significantly also the provision of knowledge to geographically, functionally, and hierarchically diverse colleagues (Reagans & McEvily, 2003). In fact, experts are more likely to read others' perspectives and problems, increasing the likelihood of boundary spanning ties. To span boundaries, individuals need to become capable of reading and understanding different perspectives, languages, and viewpoints (Bechky, 2003; Tortoriello & Krackhardt, 2010). Knowledge breadth and depth allow individuals to embed multiplex viewpoints (Beckman et al., 2014), improving the perceptual and judgmental capacities necessary for reading other perspectives and, in turn, to cross boundaries (Foss & Rodgers, 2011). Thus, expert colleagues should be better able than non-expert ones not only to establish, but also to handle crosscutting relationships.

Taken together, these considerations suggest that expert employees should be more likely than non-expert employees to have work related networks that span across intra-organizational boundaries.

Hypothesis 2. *Employees with higher levels of expertise are more likely than employees with lower levels of expertise to have boundary spanning networks.*

Important network theorizing has established a fundamental distinction between the social structure, and the content mobilized via that structure. Structure captures the configuration of social ties connecting actors in a network, such as having a given number of work relationships with certain colleagues. Content captures what actually flows through those relationships (Borgatti & Halgin, 2011). In line with this view, research on social capital has sometimes depicted the structural characteristics of a network as the infrastructure that offers the opportunity for the mobilization of resources (Adler & Kwon, 2002; Kwon & Adler, 2014). For example, different actors may be differentially induced to fully exploit ties that connect them to otherwise disconnected colleagues (Stea & Pedersen, 2017). This discussion resonates also with the debate, in the recent boundary spanning literature, highlighting the importance of taking into account the boundary spanners' relational behaviors, and particularly the active ways in which they engage in networking, provided a given structural opportunity (Monteiro & Birkinshaw, 2017).

A broader, and important implication of these perspectives is that, if network structures offer the opportunity for knowledge mobilization, then a key aspect that needs to be taken into consideration is the behavioral orientation of the social actors—that is, the extent, or even the way in which they will utilize their positions in the social space (Obstfeld, 2005; Stea & Pedersen, 2017). In contrast with the perspective that depicts networks as purely structural phenomena, an emerging stream of research argues that to fully understand the origins and impact of networks we need to capture their processual nature (Obstfeld, Borgatti, & Davis, 2014). This research perspective focuses on the social behaviors underlying network configurations. For example, Obstfeld (2005) introduced the idea of the *tertius iungens* networking behavioral orientation, that is a strategy to introduce network contacts to one another, to facilitate direct coordination and collaboration. Network behavioral orientations are fundamental micro-level behaviors that can illuminate how social networks are formed and the trajectories of their evolution (Ahuja, Soda, & Zaheer, 2012).

More specifically, in the context of relationships that link a focal employee with geographically, hierarchically, and functionally distant colleagues in the MNC, the relevant question becomes whether the focal actor has a behavioral orientation that facilitates the mobilization of resources with such a global pool of colleagues. Precisely as they are tied to individuals working in different geographic locations, organizational ranks, as well as roles along the value chain, individuals with networks that crosscut these barriers will have higher opportunities to hear different opinions and perspectives, and more broadly to tap into different knowledge pools. In order for this diversity to be put into productive use, the focal employee will need to be able to recognize and use opportunities for collaboration across diverse knowledge sources,

and behave consistently.

Employees that are able to recognize and use opportunities for collaboration between different individuals, even when these individuals have different perspectives on given issues, should thus be optimally suited to develop and maintain highly diverse networks, such as the ones that cut across intra-organizational boundaries. In their networking orientation, these individuals don't leverage informational gaps and asymmetries in the social context, but strive to connect their contacts to induce cooperation, facilitate coordination, and promote mutual adjustment (Soda, Tortoriello, & Iorio, 2018). Recently, Spiro et al. proposed a dynamic view of this type of networking orientation labeled as “matchmaking” in which “ego introduces or otherwise makes possible a tie from one alter to another” (Spiro, Acton, & Butts, 2013, p. 131). Relatedly, Kellogg (2014) conceptualized this orientation as a “connecting practice” aimed at transferring, translating, and transforming knowledge (Carlile, 2004).

In sum, employees with an open and collaborative attitude in their networking orientation should be better able than employees lacking such a collaborative networking style to establish and maintain ties that cut across intraorganizational barriers.

Hypothesis 3. *Employees with higher levels of openness and collaboration in their networking orientation are more likely to have boundary spanning networks than employees with lower levels of openness and collaboration in their networking orientation.*

2.3. The contingent effect of the formal workflow network

Taken together, the hypotheses developed so far capture the role that opportunity, capacity, and willingness have for the development of boundary spanning networks within the MNC. Yet, an important aspect to consider is whether these forces are contingent on other aspects. In particular, prior research highlights that networks and networking activities may be highly sensitive to formal and structural elements (McEvily et al., 2014). In what follows we propose that opportunity, capacity, and willingness should be considered together with the characteristics of the focal employee's formal workflow network.

Organization theory has long recognized the role of both formal and informal elements, and more recently also the “inherent interplay between formal organization and informal social structure given their common basis of interactions in organizations” (McEvily et al., 2014, p. 300). Distinct from measures capturing the patterns of informal and often spontaneously developed collaborations, we follow prior research in conceiving of the formal components of organizational architecture as networks of connections among organizational members, comprised of processes and workflows, which are enablers for accessing, exchanging or transmitting critical organizational resources such as approvals, direction, task-related information and communication (Soda & Zaheer, 2012). Thus, an employee's formal workflow network captures the formalized work processes or task interdependencies, whatever sequential or reciprocal, that an employee is involved in.

Receiving outputs from, and providing inputs to several colleagues—that is, being centrally located in the formal organizational network—increases the opportunity to interact and establish informal ties with multiple colleagues. This view is consistent with the overall idea that formal organizational structures are hierarchical social systems designed to focus the organizational members' attention (Ocasio, 1997), and to shape their social interactions (March & Simon, 1958). Thus, the formal organizational structure can be seen as a possible source of the social organization (Caimo & Lomi, 2015; McEvily et al., 2014). Based on this insight, we propose that having a mandate with a global impact, high levels of expertise, and an open and collaborative behavior should be more conducive to having boundary spanning networks when also the appropriate “formal infrastructure” is in place. In other words, employees that can benefit from a large formal workflow network should be better able to leverage on their opportunity,

capacity, and willingness so as to establish boundary spanning informal relationships. On the other hand, those employees that have the adequate task, expertise, and behavior but lack those structural opportunities that facilitate the creation of informal ties should be less able to benefit from their task, expertise, and behavior in terms of building and maintaining boundary spanning ties.

Hypothesis 4. *The effects that mandates with a global impact (H1), expertise (H2), and collaborative networking attitude (H3) have on boundary spanning are stronger for those employees that are embedded in large formal workflow networks.*

3. Methods

3.1. Research context

We tested our theory in the Supply Chain function of a global company, which we will label “BigEye” for anonymity reasons. Being one of the most critical functions for BigEye, the organization's top management was highly interested in understanding the organizational functioning “behind the chart”, particularly with respect to the patterns of informal collaborations across the employees working there. BigEye has annual net sales of more than \$9 billion, employing almost 80 thousand people, and playing the role of global leader in its market. Started in Italy as a small supplier company producing components, BigEye has rapidly grown through an ambitious acquisition strategy that allowed the company to become a top player in its industry in a relative short period of time. International expansion has developed its geographic footprint worldwide. The company's global wholesale distribution network covers more than 150 countries and is complemented by an extensive retail network of more than 7000 stores distributed globally. The company has 18 distributions centers, 12 manufacturing facilities worldwide, offices and subsidiaries in more than twenty countries, and it is present in all five continents. The Supply Chain at BigEye serves both the retail and wholesale businesses and it is globally integrated through a shared platform. Supply Chain processes are designed to link the logistics and sales centers to the production facilities and provide daily monitoring of global sales performance and inventory levels to meet local and global market demands.

One of the competitive advantages of BigEye is the vertically integrated business model built over the years, covering the entire value chain: Design, product development, manufacturing, logistics and distribution. As emphasized in the strategic planning document, BigEye delivers on its mission through its vertically integrated business model, manufacturing excellence, focus on service and geographically diversified footprint, which in turn have led to greater efficiency, flexibility and speed in supply chain and logistics.

Having facilities, plants and logistics scattered globally, BigEye has pursued a corporate strategy of integration and coordination in its global network to facilitate that best practices, knowledge and information can smoothly move from one part of the organization to another. The Supply Chain is divided into twenty distinct processes, such as S&OP & Investment Planning, Forecasting Retail, and Collaborative Planning.² Examples of jobs in Supply Chain are: Senior Planner, Planning Analysis Manager, and Product Planner.³

² Other processes are: Distribution Planning; NPI – coordination, After Sales Planning, Capacity planning, Finished Good Planning, Components Planning, Lenses Planning, Procurement, Controlling, Reporting, Projects, Scheduling & Expediting, Assortment Planning, Order Management, Lifecycle management, Intercompany Production Planning, Spare Parts Production/Sourcing Planning.

³ Other jobs are: Merchandise Planner – Sunglasses, Assistant Planner, Planning Manager, Planner – Merchandise, Merchandise Planner – Optical, Supply Chain Analyst, Allocations & Inventory Operations, SAP Master Data & Purchasing Coordinator Operations, Sales - Commercial Assistant, Planner, Industrial Planning (Geography), Sales Administration Assistant, Sales -

At the time of data collection, the Supply Chain employed 473 individuals, 57% of whom were women, scattered across five continents. Specifically, the geographical distribution of employees was the following: 35% in Europe, 35% in North America, 25% in Asia, and 5% in Latin America. We conducted several open-ended interviews with the Head of Supply Chain and the top management team, which helped us better understand the organization and its key processes.

3.2. Network data

Network analysis is used to code the dependent variable based on three network adjacency matrices mapping the hierarchical, intra-functional, and geographical distances between each dyad. The same applies to the calculation of the size of each employee's workflow, and friendship networks. Those measures as well were coded according to standard network analytical practices. These processes are complemented by matching them with statistical analyses where the hypothesized effects of our predictors on the dependent variable are tested on Ordinary Least Square (OLS) Regression models.

Specifically, a web-based network survey was administered to 473 employees of the Supply Chain at BigEye. The survey package included a personalized invitation e-mail containing a unique code to enter the system, and a cover letter from the Head of Supply Chain introducing the study. Because of the multicultural nature of the organization, employees were given the opportunity to select their favorite idiom among seven different languages by clicking on national flags in the survey homepage. In order to ensure confidentiality, the surveys were administered through a Business School on-line platform and names were replaced by unique number identifiers. One week after the employees received the initial survey a reminder email was sent. The survey remained open for three weeks. The survey took an average of twenty-five minutes to be completed and we obtained a response rate of 60%.

The network data were collected using classic sociometric techniques (Wasserman & Faust, 1994). Upon reading the instructions of the survey, respondents were asked to fill out a bar with names of people to whom they turned for advice on work-related issues. Thus, the network metric was based on people's individual outgoing choices, which have the advantage to be largely under the control of the actor. As Borgatti and Cross (2003) point-out these measures are largely reliable when researchers follow appropriate procedures to help individuals recalling their social structure. In this regard, the survey was pre-tested with a pilot study involving a limited number of people from the upper-level management team to create network questions as specific as possible that elicited long-term patterns of interaction rather than one-time events (Borgatti & Cross, 2003; Freeman, Romney, & Freeman, 1987). In addition, as soon as the respondents started typing the first letters of their contacts' name or surname, the on-line survey tool suggested names of Supply Chain employees matching with the letters inserted, thus easing the "alters" selection task. Hence, we both avoid typos and, at the same time, potential intricacies arising from a roster with hundreds of names.

We examined our survey data for the risk of non-response bias. First, we discussed in a specific meeting the results and demographic breakdown of respondents (e.g., by age, education, and tenure) with the head of Supply Chain and his direct reports; all assured us there were no visible biases differentiating those responding to the survey from the overall distribution of Supply Chain organizational members. Furthermore, we conducted a wave analysis comparing the demographic variables for early (1st week) versus late (3rd week) respondents

(Rogelberg & Stanton, 2007). The assumption is that the group of late respondents will be more similar to the non-responding group than to the group of early respondents. An analysis of variance (ANOVA) of the difference in means for the two groups for the demographic variables showed that the hypotheses of differences in means could all be rejected (F -values < 2). In addition, we obtained archival data from other sources like detailed demographic data from the HR-department and data on individuals' performance assessed by the managers as the company has an extended performance evaluation system.

After the data cleaning procedure and imputation of a few missing values (where we had holes in an otherwise complete data string) we were able to use data on 275 employees in all our analyses, representing 58 percent of all employees in Supply Chain at the time of the survey.

3.3. Measures

The included variables are calculated based on five different sources: Network data based on ego's and alters' reports (advice network, workflow network, friendship network, brokerage), self-reported measures based on employees' reports of own behavior (networking orientation, face-to-face communication), alters' assessment of ego's characteristics (expertise), managers' assessment of employee's performance (task performance), and archival data on demographics and task characteristics (global impact, upstream activity, organizational rank, education, gender). By relying on these multiple sources we are limiting common method bias concerns that generally arise when respondents provide self-reported data on multiple variables (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003; Podsakoff, MacKenzie, & Podsakoff, 2012).

Self-reported measures have well-known weaknesses—for example, individuals may have biased perceptions of own behavior—yet, they remain an accepted way of capturing perceptions and behaviors among employees (Howard, 1994). In particular, employees may be better aware of the more subtle things that they do in their jobs (Shalley, Gilson, & Blum, 2009) and, for this reason, self-reports have been used to capture constructs such as individual knowledge mobilization (Reinholt et al., 2011), and individual creativity (Janssen, 2000; Shalley et al., 2009). Based on the same intuition, we use self-reported measures to capture those constructs for which ego is arguably the better informed actor—that is, ego's own networking orientations, as well as the nature of work-related communication carried out by ego.

Nevertheless, social desirability and other respondent biases such as common method bias are obvious limitations of self-reported measures. However, the questionnaire used different scales, some of which were reversed, diminishing the risk of bias (Podsakoff et al., 2003). Furthermore, we performed a number of statistical analyses to assess the severity of respondent biases. First, we conducted a Harman's one-factor test on the items, and the results indicated that common methods bias was not a major issue. Multiple factors were detected, and the variance did not merely stem from the first factors (Podsakoff & Organ, 1986). In fact, the 13 variables included in the model (listed in Table 1) form several factors with an eigenvalue > 1, and the first two factors only captured 20% and 12% of the total variance, respectively. Second, following Podsakoff et al. (2003), we ran a partial least square (PLS) model, including a common method factor whose items encompassed all of the construct's items. This PLS model provided information on each item's variances, substantively explained by the constructs and the common method factor, respectively. The average of substantive variance explained by the constructs was between 0.54 and 0.74, whereas the average method variance was around 0.02 for all items. The ratio of substantive variance to method variance was therefore very high, making us confident that our data does not suffer from major respondent biases.

These statistical tests do not eliminate the threat of respondent biases, yet they suggest that our results are not driven predominantly by common method variance. Lastly, while management urged employees to respond to the survey, employees were ensured that individual

(footnote continued)

Commercial Assistant, Supply Chain - Demand & Distribution Planning Manager, Planning GMO - Product & Buying Specialist, Planner Sun & Accessories, Product Frames Manager.

Table 1
Correlation Matrix.

	1	2	3	4	5	6	7	8	9	10	11	12	13
1. Boundary spanning network	1												
2. Global impact	0.20	1											
3. Expert	0.29	0.17	1										
4. Collaborative networking orientation	0.26	0.06	0.14	1									
5. Size of workflow network	0.16	0.11	0.47	0.16	1								
6. Size of friendship network	−0.05	0.04	0.18	−0.02	0.32	1							
7. Brokerage	0.05	−0.05	0.27	0.08	0.30	0.15	1						
8. Face-to-face communication	−0.04	0.11	0.06	0.12	0.12	0.11	0.10	1					
9. Task performance	0.11	−0.01	0.11	0.13	0.05	0.05	0.07	0.09	1				
10. Organizational rank	−0.28	−0.16	−0.36	−0.23	−0.46	−0.17	−0.13	−0.11	−0.01	1			
11. Upstream activity	0.02	−0.14	0.02	−0.03	−0.20	−0.21	−0.06	−0.19	−0.02	0.18	1		
12. Education	0.03	0.05	0.10	−0.15	0.04	−0.06	−0.01	−0.05	−0.06	0.12	0.19	1	
13. Gender	0.20	0.07	0.20	0.22	0.07	−0.01	0.03	−0.04	0.09	−0.19	0.12	−0.04	1
Mean	0	0.19	10.52	4.10	5.31	1.75	0.65	0.63	3.49	3.07	0.39	2.44	0.43
St dev	2.03	0.39	15.04	0.46	4.84	1.39	0.24	0.48	0.82	0.80	0.49	1.50	0.50
Min	−2.77	0	0	2.49	0	1	−0.12	0	1	0	0	1	0
Max	5.95	1	84	5	24	11	1	1	5	4	1	5	1

All coefficients above |0.12| are significant at 5% level of significance.

Mean, st. dev, min and max values is before standardization (except for our dependent variable that are standardized from the outset).

responses would not be disclosed within BigEye. The survey was hosted on a server outside the firm and only aggregate-level data was reported back to the firm. Employees' knowledge of this further reduces the likelihood of biased responses (Podsakoff et al., 2003, 2012).

3.3.1. Dependent variable

Boundary spanning network. We asked each respondent to identify those colleagues “to whom you turn to seek advice on work-related issues”. The item was chosen in line with the idea that advice ties provide an important basis for understanding key aspects of knowledge mobilization in organizations (Podolny & Baron, 1997). Based on employees' responses, we created an adjacency matrix of network relations (relation or not = 1/0) among the employees who either responded themselves, or were listed by their colleagues. Since all advice ties might be vehicles for knowledge mobilization, we symmetrized the matrix following the rule that when either member of a pair nominated the other, the pair was considered to have a tie (Mehra, Kilduff, & Brass, 2001; Reagans & McEvily, 2003).

Consistent with the insights developed in the IB literature, the boundary spanning measure is calculated across the three dimensions of geography (Cramton, 2001; Espinosa et al., 2003), hierarchy (Reagans et al., 2004; Zenger & Lawrence, 1989), and function (Bartlett et al., 2003; Puranam et al., 2012) where, based on the adjacency matrix, we constructed three additional matrices depicting whether each possible dyad was spanning across geographical, hierarchical, and intra-functional boundaries. Specifically, the geographical dimension is captured by the share of the total advice ties that link ego to alters located in countries different than ego's own country (mean = 16 percent). The hierarchical dimension is calculated as the share of alters in the ego network that are on a different organizational rank (either higher, or lower) than ego (mean = 49 percent). The intra-functional dimension captures the share of alters that are involved in different—upstream versus downstream—activities within the supply chain function than ego (mean = 12 percent). These three ratios—which span from 0 to 100 percent—are first standardized (mean = 0, standard deviation = 1) in order to give them equal weight, and then added together in order to create a composite index of the extent to which each employee's ego network crosses intraorganizational boundaries. Thus, this measure does not require that all types of boundaries are crossed simultaneously, but it allows for, and empirically captures also that possibility. The skewness of this variable is 0.59 and the kurtosis is −0.19, which is within the acceptable range for skewness and kurtosis (of ± 1), and is also reflected in the normality test of the variable.

3.3.2. Independent variables

Global impact is a dummy variable provided by BigEye's HR Department indicating whether the focal employee has a mandate that the organization regards as having a global impact (yes = 1, no = 0). That is, whether ego's assignments include responsibilities that apply on a global scale, such as conducting projects that affect practices at the global level or working for services which are delivered at the global level. Of all respondents, 19 percent have mandates with global impact.

Expert. Respondents were asked for the twenty different processes listed in the Research Context section: “Who do you consider to be the two greatest experts, inside the Supply Chain community, for each of the following processes?” This question generated a list of experts for each of the twenty processes. For each employee, we counted the number of times that he or she had been mentioned as an expert across the twenty processes by her colleagues, making it a multi-rater measure of ego's expertise. Out of the 279 individuals in our analysis, 80 percent have been mentioned as experts at least once, and 50 percent are mentioned more than 3 times.

Collaborative networking orientation. Respondents were also asked a number of questions about their broader networking orientations. The following three questions, adapted from Obstfeld (2005), and measured on a 5-point scale (1 = strongly disagree, 5 = strongly agree), were used to form a composite variable capturing the extent to which ego is quick in recognizing and using opportunities for collaboration between different individuals: “I see opportunity to act as a bridge conveying information from one person to another”, “I point out common ground shared by people who have a different perspective on an issue” and “I see opportunity for collaborating between people”. The means for the three questions on the 5-point scale are 3.9, 4.0, and 4.4, respectively. The obtained Cronbach's alpha value for this construct was 0.73, and the composite reliability was 0.65, with an AVE-value of 0.74. These measures suggest the construct is highly reliable, and characterized by convergent validity.

Size of workflow network captures the size (Freeman, 1979) of ego's formal workflow network (as reported by alters), and is calculated by first asking to all individuals in the network to identify those colleagues that “provide you with workflow inputs” and “to whom you provide your workflow outputs,” and then summing the number of alters that mention ego as someone to whom they provide workflow outputs, plus the number of alters that mention ego as someone from whom they receive workflow inputs. The variable has a median of 3.5, and is used as a control in Model 1, and to split the sample between employees with small (≤ 3.5) and large formal workflow networks (> 3.5) in Model 2.

3.3.3. Control variables

The models include a number of control variables capturing structural (size of friendship network, brokerage), relational (face-to-face communication), and individual (task performance, organizational rank, upstream activity, education, gender) characteristics of the focal employee.

Size of friendship network counts the number of reciprocated friendship ties in each ego network, and is calculated by counting for each ego the number of relationships where both ego considers alter, and alter considers ego “to be a good friend”. This variable controls out the possibility that boundary-crossing might be facilitated by feelings of friendship (Kilduff, 2004).

Brokerage was calculated in the structural holes routine in UCINET (Borgatti, Everett, & Freeman, 2002), where we specified the measure as the additive inverse of Burt's (1992) constraint measure. The constraint measure is a function of the network's density, size, and hierarchy and is meant to capture the extent to which the focal actor's network lacks structural holes. This operationalization of the brokerage measure is consistent with a long research tradition (Burt, 1992) as well as with more recent contributions building on that tradition (Carnabuci & Dioszegi, 2015; Soda et al., 2019). Constraint varies from 0 to 1, and, as is customary, we used 1 minus constraint to directly measure a focal actor's involvement in a brokering position.

Face-to-face communication captures whether ego enacts her network ties directly (i.e., via face-to-face communication) or indirectly (e.g., via mail, or videoconferencing). Respondents were asked “what are the ways in which you manage your network inside the Supply Chain community?” and then a number of communication mechanisms were provided. The three options: “meetings”, “informal occasions at work” and “informal occasions (outside) work” were collapsed into a measure of face-to-face communication ($= 1$), all other communication options (mail, video conference, Skype, Facebook, LinkedIn) were assigned a value of 0, as they indicated the focal employee used more indirect means of communication. Of all respondents, 53 percent were mainly using face-to-face communication, and 47 percent mainly used indirect means of communication. The variable is used to tease out the potential effect that different communication media might have on how individuals enact their networks.

Task performance is the supervisor's assessment of ego's work performance. BigEye has designed for the Supply Chain a management-by-objectives evaluation process where employee task performance is rated by each employee's supervisor as “very low”, “low”, “fair”, “good”, or “outstanding” (scale from 1 to 5). The evaluation process is conducted annually, and we obtained directly from BigEye's HR office the performance assessment for all the participants in the study. The mean value on the 5-point scale is 3.5, and the median is 3, indicating that employees are somewhat evenly spread on the scale, but with a bulk having good performance. This variable controls out the potential confounding explanation that ego's boundary crossing activity might be related to ego's task performance.

Organizational rank, as provided by the HR-department, measures the focal employee's hierarchical level on a scale from 0 (high rank) to 4 (low rank). The median of this variable is 3. This variable controls out the possibility that individuals at different hierarchical levels have different opportunities for having cross-boundary relationships.

Upstream activity is a dummy variable (1 = upstream, 0 = downstream) controlling for whether the focal actor is involved in activities that are upstream (e.g., procurement) or downstream (e.g., sales) in the value chain. Of all respondents, 39 percent work in upstream activities, and 61 percent in downstream activities. This variable controls for the possibility that different activities in the value chain imply different needs or opportunities for creating cross-boundary relationships.

Education, as reported by the focal employee, is measured on a four points scale (1 = Bachelor or lower, 2 = Master, 3 = MBA, and 4 = PhD), and controls for the possibility that different educational levels

imply different abilities or opportunities to engage in boundary-crossing relationships.

Gender is a dummy variable (male = 1 and female = 0) provided by the HR-department and aimed at controlling for the potential effect of gender on the engagement in boundary-crossing ties.

4. Analysis and results

The correlation matrix for all variables including mean values, standard deviation, minimum and maximum values are shown in Table 1. It is noticeable that all three hypothesized variables (global impact, expert and collaborative networking orientation) have significant and positive bivariate correlations with the boundary spanning index (from 0.20 to 0.29).

None of the independent variables have correlations that indicate problems of multicollinearity. The highest bivariate correlations are between the size of the employee's workflow network and her expertise ($= 0.47$), and between the size of the employee's workflow network and her organizational rank ($= -0.46$, where highest rank = 0 and lowest rank = 4 in our scale, hence the negative sign in the correlation). Both correlations are not surprising. Experts as well as highly ranked employees should be more likely than non-experts or lower ranked employees to be involved in several formalized work processes or task interdependencies. However, we conducted all models with and without the highly correlated variables, and results did not change. All variables in our models are standardized (mean = 0 and standard deviation = 1), allowing us to compare the size of the coefficients across the different variables, since they are all on the same (standardized) scale.

Our data might potentially suffer from selection bias, as respondents have not been collected in a randomized way. Therefore, we conducted a Heckman-model in order to test the severity of the selection bias in our data. The first stage consists of an equation where the (binary) dependent variable express whether the employee has a boundary crossing index of zero (dummy = 0, $n = 27$) or more (dummy = 1, $n = 248$). The second stage consists of an equation explaining the level of the boundary crossing index. If present, a selection bias would be reflected in a significant correlation between the error terms in the two equations, as this implies common observables between the selection and levels of our dependent variable. However, in our case we obtain an insignificant correlation of -0.04 with $p = 0.90$, suggesting that the data does not suffer from selection bias. This is also consistent with the fact that our data has been collected from multiple sources, and includes variables that are not determined by the focal employee (such as the focal employee's expertise as assessed by her colleagues, or the employee's global impact, which is normally granted by the employee's manager).

As our dependent variable is a normally distributed continuous variable and the data does not suffer from selection bias, we can test our hypotheses in an Ordinary Least Square (OLS) Regression model. In addition, when running the OLS models, we also conducted tests for non-normality of the error terms, heteroscedasticity, and examined the residual plots.

Table 2 includes the OLS models. Model 1 is our main model conducted on the entire population, and is used to test Hypotheses 1, 2, and 3. Model 2 is used to test Hypothesis 4, and splits the sample in employees with large versus small workflow networks.

Model 1 explains 21 percent of the variation in the boundary crossing index. The three hypothesized variables, global impact, expert, and collaborative networking orientation are all positive and significant, as hypothesized. As we can directly compare the coefficients of the standardized variables, we can see that employees recognized as experts by their colleagues ($\beta = 0.44$, $p = 0.004$) obtain the largest coefficient, while the coefficients for collaborative networking behavior ($\beta = 0.38$, $p = 0.003$) and global impact ($\beta = 0.29$, $p = 0.02$) are somewhat lower. Taken together, Model 1 confirms Hypotheses 1, 2,

Table 2

OLS Regression models with the share of boundary-crossing ties as dependent variable (std. errors in parentheses).

	Model 1		Model 2			
	All observations		Small workflow		Large workflow	
Global impact	0.29*	(0.12)	−0.03	(0.21)	0.54***	(0.12)
Expert	0.44**	(0.16)	0.34	(0.51)	0.35**	(0.13)
Collaborative networking orientation	0.38**	(0.12)	0.51**	(0.19)	0.16	(0.14)
Size of workflow network	−0.16	(0.17)				
Size of friendship network	−0.17	(0.12)	−0.67*	(0.28)	−0.08	(0.11)
Brokerage	−0.01	(0.12)	−0.04	(0.15)	0.10	(0.27)
Face-to-face communication	−0.18	(0.12)	−0.01	(0.19)	−0.28*	(0.14)
Task performance	0.13	(0.11)	0.34*	(0.18)	−0.03	(0.14)
Organizational rank	−0.39**	(0.13)	−0.53*	(0.24)	−0.27*	(0.14)
Upstream activity	0.02	(0.12)	−0.21	(0.20)	0.27	(0.16)
Education	0.10	(0.12)	0.35	(0.20)	−0.11	(0.13)
Gender	0.13	(0.12)	0.01	(0.19)	0.33*	(0.14)
Intercept	0.01	(0.11)	0.17	(0.30)	−0.06	(0.15)
F-value	5.75***		2.31*		8.23***	
R2	0.21		0.17		0.42	
N	275		136		139	

*, **, and *** indicate 0.05, 0.01, and 0.001 percent level of significance, respectively.

and 3 proposing that employees having mandates with a global impact, high expertise, and a collaborative networking orientation will be more likely than employees without those characteristics to have networks rich in boundary spanning ties.

Among the control variables, only organizational rank has a significant effect. The fact that individuals at higher ranks have higher share of cross-boundary ties seems intuitive. High rank employees normally have more opportunities as well as obligations to engage in relationships that cross intra-organizational boundaries.

In Model 2 we conduct a split sample analysis dividing the population between employees with small (below median, $n = 136$) versus large (above median, $n = 139$) formal workflow networks. The results obtained in Model 1 mainly hold for those employees with large workflow networks, where no less than 42% of the variation in the boundary spanning index is explained by the model. Specifically, having a mandate with a global impact ($\beta = 0.54$, $p < 0.0001$) and being perceived as an expert ($\beta = 0.35$, $p = 0.002$) has a positive and significant effect only for employees with large formal workflow network. However, the same does not apply to the collaborative orientation variable, where in fact, the effect remains positive and significant only for those employees with small workflow networks. Taken together, these results suggest that Hypothesis 4 is confirmed for global impact, and expertise, but not for networking behavior. This may be suggestive of the fact that collaborative networking behaviors may compensate for the lack of formal drivers or expertise in the formation of boundary spanning ties.

4.1. Additional analyses

Lastly, while our dependent variable includes geographical, hierarchical, and intra-functional distances precisely in order to offer a comprehensive boundary-crossing index that simultaneously touches upon these three important dimensions identified by prior research, we also conducted our models for each dimension separately, in order to further explore whether there are systematic differences in how employee's mandates, skills, and behavioral orientations affect specific types of boundary spanning ties.

Table 3 includes the OLS models testing separately the effects of our

Table 3

OLS Regression models with the three dimensions of boundary spanning modeled separately (std. errors in parentheses).

	Model 3		Model 4		Model 5	
	Geography		Function		Hierarchy	
Global impact	0.01	(0.06)	0.09	(0.06)	0.20***	(0.06)
Expert	0.16*	(0.08)	0.21**	(0.08)	0.06	(0.08)
Collaborative networking orientation	0.18**	(0.06)	0.07	(0.06)	0.13*	(0.06)
Size of workflow network	−0.09	(0.09)	−0.21*	(0.09)	0.13	(0.08)
Size of friendship network	−0.03	(0.06)	−0.05	(0.06)	−0.08	(0.06)
Brokerage	0.01	(0.06)	0.05	(0.06)	−0.06	(0.06)
Face-to-face communication	−0.03	(0.06)	−0.05	(0.06)	−0.10	(0.06)
Task performance	−0.04	(0.06)	0.13*	(0.06)	0.06	(0.06)
Organizational rank	−0.10	(0.07)	−0.14*	(0.07)	−0.15*	(0.07)
Upstream activity	−0.25***	(0.06)	0.13*	(0.06)	0.14*	(0.06)
Education	0.08	(0.06)	0.02	(0.06)	−0.01	(0.06)
Gender	0.03	(0.06)	0.09	(0.06)	0.01	(0.06)
Intercept	0.01	(0.06)	0.01	(0.06)	0.01	(0.06)
F-value	3.25***		3.72***		4.74***	
R2	0.13		0.15		0.18	
N	275		275		275	

*, **, and *** indicate 0.05, 0.01, and 0.001 percent level of significance, respectively.

explanatory and control variables on the spanning of geographical (Model 3), intra-functional (Model 4), and hierarchical (Model 5) boundaries.

While these additional analyses show that our main results are not predominantly driven by one dimension in the sense that boundary spanning happens on all the three dimensions, they also show that, although mandates, skills, and behavioral orientations all present positive coefficients, the magnitude of their effects differs significantly, depending on the boundaries spanned. Specifically, expertise strongly affects boundary spanning across intra-functional ($\beta = 0.21$, $p = 0.009$) and geographical ($\beta = 0.16$, $p = 0.05$) boundaries, having a collaborative orientation drives the spanning of geographical ($\beta = 0.18$, $p = 0.005$) and hierarchical ($\beta = 0.13$, $p = 0.03$) distances, and having a mandate with a global impact predominantly drives the spanning of hierarchical ($\beta = 0.20$, $p = 0.001$) distances.

These results offer additional insights and suggestions for future research, which we unfold in the discussion section.

4.2. Robustness checks

We conducted a number of robustness checks in order to confirm the validity of our findings. First, we tested our models on alternative operationalizations of the dependent variable. In particular, we used cities instead of countries to measure the ties that crosscut geographical boundaries, and the magnitude of the hierarchical gap (in terms of how many levels separated each ego-alter dyad) to measure the ties that crosscut organizational ranks. The results were unchanged.

Second, we conducted all models excluding in turn the more highly correlated variables, in order to detect potential problems of multicollinearity. Also in this case, the results did not change, supporting our confidence in the results.

5. Discussion

While it is widely accepted that boundary spanning ties are very beneficial to MNCs (Birkinshaw et al., 2017; Kluter & Monteiro, 2017; Monteiro & Birkinshaw, 2017; Schotter et al., 2017), research on social capital has also raised awareness about the potentially problematic nature of ties that cut across different organizational boundaries (Caimo

& Lomi, 2015; Soda et al., 2019; Stea & Pedersen, 2017). On the one hand, boundary spanning ties are likely to tap into different knowledge structures which facilitate innovation (Cohen & Levinthal, 1990), lubricate the workings of costly and complex formal organizational processes (McEvily et al., 2014), and have well-documented implications for organizational performance (Burt, 2004; Reagans & McEvily, 2003; Reagans & Zuckerman, 2001). On the other hand, ties linking to different knowledge pools often imply high levels of uncertainty, ambiguity, and difficulty in communications (Tushman & Scanlan, 1981), and are for these reasons more difficult both to establish (McPherson et al., 2001), and to handle (Reagans & McEvily, 2003; Stea & Pedersen, 2017; Tortoriello et al., 2012). The implicit tension between MNCs' need for boundary spanners, and the inherent difficulties associated with boundary spanning ties in the workplace has triggered the question: Which employees are more likely to develop work related networks that crosscut intraorganizational barriers?

Our answer to these questions builds on opportunity-capacity-willingness theories of behavior (Blumberg & Pringle, 1982; Roberts & Beamish, 2017), and shows that the employees that are more likely to be boundary spanners are those having work mandates with a global impact, high levels of expertise, and a collaborative networking orientation. We also find that these effects are sensitive to the size of employees' formal workflow networks.

In order to further investigate the nature of these results, we disaggregated our dependent variable into the three key dimensions of geographical, intra-functional and hierarchical boundary spanning, and explored the specific impact of our explanatory variables on each dimension separately. By doing so, we obtained additional interesting insights.

In particular, when looking at the coefficients in the model, it seems that mandate, expertise, and networking orientations have stronger effects on the spanning of different types boundaries. Specifically, expertise is particularly important in affecting the spanning of intra-functional boundaries, mandates with a global impact are most conducive to the spanning of hierarchical boundaries, and a collaborative networking orientation is important for developing ties that crosscut geographical barriers.

These results are interesting as, while confirming the broader intuition that opportunity, capacity and willingness are key prerequisites for boundary spanning in global organizations, they also provide initial evidence on which aspects matters the most for the spanning of specific boundaries, while controlling out for potential confounding factors.

In particular, the fact that having an assignment identified by the organization as one with a global impact makes it easier to reach out to colleagues across hierarchical levels, might indicate that these mandates project credibility on the focal employee, paving the way for facilitated access to people at different hierarchical levels. The workings of expertise seem to be quite distinct. In fact, expertise has no sizeable effect on the spanning of hierarchical boundaries but has the strongest effect on intra-functional boundary spanning. That may be suggestive of the fact that expertise is more of a technical enabler of boundary spanning, where the effects for boundary spanning of being an expert are mostly visible at the more operational level, where experts are of key importance for coordinating activities all along the value chain. Lastly, having a collaborative networking orientation particularly affects the spanning of geographical distances. This result may indicate that these more social, and relational skills are particularly meaningful for handling work relationships that often require a good appreciation of potentially subtle cultural or environmental differences.

Taken together, the intuitions developed in this paper extend existing research in a number of ways. First, an important discussion in IB research has developed around the importance of boundary spanners (Minbaeva & Santangelo, 2017; Monteiro & Birkinshaw, 2017; Schotter et al., 2017), and more generally about the value of mobilizing knowledge resources across those barriers that are to some extent a by-product of the international nature of the activities carried out in MNCs

(Doz et al., 2001; Kogut & Zander, 1993). Recent contributions have also highlighted a lack of knowledge about how boundary spanning works (Birkinshaw et al., 2017; Monteiro & Birkinshaw, 2017), and more generally about the factors that affect the effectiveness of boundary spanning (Schotter et al., 2017). Using theoretical and methodological insights from network theory, our contribution to this discussion is to show that networks of boundary spanning ties are more likely to be developed by employees with specific tasks, abilities, and behaviors. In that sense, our findings expand understanding of those micro-level mechanisms and processes that may overcome some of the barriers to knowledge mobilization that are typical of MNCs (Caimo & Lomi, 2015).

Second, prior research has produced considerable insight into how knowledge is mobilized between the subsidiaries of MNCs (Foss & Pedersen, 2002; Gupta & Govindarajan, 2000), yet much less is known about knowledge mobilization across the individual organizational members of MNCs. Consistently, research has called for greater insight into the microfoundations of the knowledge-based processes in MNCs (Foss & Pedersen, 2004; Foss et al., 2010; Minbaeva & Santangelo, 2017). In particular, while acknowledging the importance of geographic, functional, and hierarchical boundaries in MNCs, most analyses of individual level, intra-organizational boundary spanning focus on one specific type of boundaries at a time (Minbaeva & Santangelo, 2017; Schotter et al., 2017). Different from this approach, we test the idea that many of the factors that can be associated with boundary spanning may simultaneously affect the spanning of hierarchies, intra-functional domains, and geographic territories. Our contribution in this sense is to offer a more comprehensive account of boundary spanning in global organizations.

Third, our findings also advance research on social networks. While network research has had a long-standing interest in the structural properties of individuals' networks such as density, openness, or centrality (Borgatti & Halgin, 2011; Borgatti, Mehra, Brass, & Labianca, 2009), a much less explored, and yet fundamental property of networks is their enactment, which can also be expressed in terms of how individuals use their networks (Soda et al., 2019; Stea & Pedersen, 2017; Stea, Pedersen, & Foss, 2017). This debate resonates with the discussion, in the boundary spanning literature, on the tendency to conflate "position" occupied and "role" performed by the boundary spanners (Birkinshaw et al., 2017; Monteiro & Birkinshaw, 2017; Schotter et al., 2017). As pointed out by Schotter et al. (2017), one body of literature tends to present boundary spanning as a more institutionalized function, which is facilitated by organizational structures, routines, or assignments (Dunning, 1998). On the other hand, the alternative and quite different approach, tends to consider boundary spanning from a more behavioral angle, treating it as something that mostly rests on employees' motivation, ability, and behaviors (Mudambi & Swift, 2009; Schotter & Beamish, 2011). These bodies of literatures have rarely been integrated (Schotter et al., 2017). A contribution of our model, relative to these approaches, is to simultaneously account for the effect of the more institutionalized factors such as having a mandate with global impact, or a large workflow, as well as the more behavioral ones, such as having the right networking orientation, or level of expertise.

5.1. Managerial implications

Given the importance of knowledge mobilization and innovation for MNCs, confirming the validity, and defining enhancers of the relationship between employees' tasks, expertise, and behaviors and the presence of boundary spanning ties is important to practitioners. Managers make choices that impact the creation, modification, and evolution of employees' networks, and many of these choices may directly relate to key variables of interest to this study. For example, managers make decisions on projects, links between units, and task assignments that may influence the extent to which employees have assignments with a global impact for the organization. Similarly,

managers can favor employee development and training programs that facilitate the accumulation of expertise. Further, managers can play a fundamental role in promoting the establishment of an open and collaborative culture in the workplace, which may promote the ability and willingness to recognize and use opportunities for collaboration. Thus, a first practical implication of this study is to alert managers that they should be cognizant of employees' tasks, expertise, and network behaviors for cross-border knowledge mobilization, and that they should factor this understanding into their decision making.

A second practical implication can be drawn by looking at our findings from the employee's viewpoint. While employees are probably less able than their managers to influence the larger social context where they operate, they can make choices that influence the structure of their networks as well as their own behaviors. For example, by committing to tasks with a global reach, employees may increase their chances to develop boundary spanning ties. In this sense, the simple and yet powerful implication of our study for employees is to be aware of the fact that personal development, and networking behaviors are necessary for them to increase the chance to develop boundary spanning ties.

In conclusion, a better understanding of how the role of opportunity, capacity and willingness for the occupation of boundary spanning networks in the MNC not only enriches our theoretical understanding of these important phenomena, but also provides managers as well as employees with refined guidance for value creation.

5.2. Limitations and future research

While this study was conducted on a rigorous survey- and network-based methodology, the data are not free from limitations. These limitations imply caution in interpreting our results, but also offer clear directions for future research.

First, as we rely on cross-sectional data, the direction of causality in our model cannot be fully ascertained. That is, while we argue in the direction from task, expertise, and behavioral orientations to network characteristics it is also possible to hypothesize alternative causal explanations. For example, our boundary-crossing network measure could reflect pre-existing individual characteristics, because people who have very global networks may be more likely to be perceived as experts by their peers. However, our arguments run in the opposite direction, consistent with abundant evidence that networks are to a certain extent malleable structures influenced by formal opportunities, as well as personal, contextual, and socio-structural factors (Ahuja et al., 2012). Despite our results' consistency with the theoretical reasoning, our study's cross-sectional design did not allow us to rule out alternative explanations. Additional research based on experimental or longitudinal data will be needed to confirm the direction of causality our model proposed.

Additionally, it is important to be cautious in generalizing our results. Specifically, the research was conducted in the Supply Chain group of a large, global MNC. While this MNC does not operate differently from most other large, distributed MNCs, our theory should be examined in other functions, as well as in other organizations.

Lastly, firm-level contextual contingencies are black-boxed in our model. While our study is based on a single organization with a rather consistent culture for knowledge mobilization and interpersonal collaboration, existing research has stressed the importance of company-level contingencies for network effects (Xiao & Tsui, 2007), and particularly of variables that are supportive of knowledge mobilization (Foss, Pedersen, Reinholt Fosgaard, & Stea, 2015) and interpersonal collaboration (Soda et al., 2019). These contextual variables should be included, particularly when scaling our analyses up so as to include data from multiple organizations.

References

- Adler, P. S., & Kwon, S.-W. (2002). Social capital: Prospects for a new concept. *Academy of Management Review*, 27(1), 17–40.
- Ahuja, G., Soda, G., & Zaheer, A. (2012). The genesis and dynamics of organizational networks. *Organization Science*, 23(2), 434–448.
- Bartlett, C. A., Ghoshal, S., & Birkinshaw, J. (2003). *Transnational management: Text, cases and readings in cross border management*. Boston: McGraw-Hill.
- Bechky, B. A. (2003). Sharing meaning across occupational communities: The transformation of understanding on a production floor. *Organization Science*, 14(3), 312–330.
- Beckman, C. M., Schoonhoven, C. B., Rottner, R. M., & Kim, S.-J. (2014). Relational pluralism in de novo organizations: Boards of directors as bridges or barriers to diverse alliance portfolios? *The Academy of Management Journal*, 57(2), 460–483.
- Birkinshaw, J., Ambos, T. C., & Bouquet, C. (2017). Boundary spanning activities of corporate HQ executives insights from a longitudinal study. *Journal of Management Studies*, 54(4), 422–454.
- Blumberg, M., & Pringle, C. D. (1982). The missing opportunity in organizational research: Some implications for a theory of work performance. *The Academy of Management Review*, 7(4), 560–569.
- Borgatti, S. P., & Cross, R. (2003). A relational view of information seeking and learning in social networks. *Management Science*, 49(4), 432–445.
- Borgatti, S. P., & Foster, P. C. (2003). The network paradigm in organizational research: A review and typology. *Journal of Management*, 29(6), 991–1013.
- Borgatti, S. P., & Halgin, D. S. (2011). On network theory. *Organization Science*, 22(5), 1168–1181.
- Borgatti, S. P., Everett, M. G., & Freeman, L. C. (2002). *Ucinet for Windows: Software for social network analysis*. Harvard, MA: Analytic Technologies.
- Borgatti, S. P., Mehra, A., Brass, D. J., & Labianca, G. (2009). Network analysis in the social sciences. *Science*, 323(5916), 892–895.
- Bouquet, C., & Birkinshaw, J. (2008). Weight versus voice: How foreign subsidiaries gain attention from corporate headquarters. *The Academy of Management Journal*, 51(3), 577–601.
- Brass, D. J., Galaskiewicz, J., Greve, H. R., & Tsai, W. (2004). Taking stock of networks and organizations: A multilevel perspective. *The Academy of Management Journal*, 47(6), 795–817.
- Burt, R. S. (1992). *Structural holes: The social structure of competition*. Cambridge, MA: Cambridge, MA.
- Burt, R. S. (2000). Decay functions. *Social Networks*, 22(1), 1–28.
- Burt, R. S. (2004). Structural holes and good ideas. *The American Journal of Sociology*, 110(2), 349–399.
- Caimo, A., & Lomi, A. (2015). Knowledge sharing in organizations: A bayesian analysis of the role of reciprocity and formal structure. *Journal of Management Studies*, 41(2), 665–691.
- Cannella, A. A., & McFadyen, M. A. (2016). Changing the exchange: The dynamics of knowledge worker ego networks. *Journal of Management*, 42(4), 1005–1029.
- Carlile, P. R. (2004). Transferring, translating, and transforming: An integrative framework for managing knowledge across boundaries. *Organization Science*, 15(5), 555–568.
- Carnabuci, G., & Dioszegi, B. (2015). Social networks, cognitive style, and innovative performance: A contingency perspective. *The Academy of Management Journal*, 58(3), 881–905.
- Cohen, W. M., & Levinthal, D. A. (1990). Absorptive capacity: A new perspective on learning and innovation. *Administrative Science Quarterly*, 35(1), 128–152.
- Coleman, J. S. (1988). Social capital in the creation of human capital. *The American Journal of Sociology*, 94, 95–120.
- Cramton, C. D. (2001). The mutual knowledge problem and its consequences for dispersed collaboration. *Organization Science*, 12(3), 346–371.
- Dasí, Á., Pedersen, T., Gooderham, P. N., Elter, F., & Hildrum, J. (2017). The effect of organizational separation on individuals' knowledge sharing in MNCs. *Journal of World Business*, 52(3), 431–446.
- Doz, Y., Santos, J., & Williamson, P. (2001). *From global to metanational: How companies win in the knowledge economy*. Boston, MA: Harvard Business School Press.
- Dunning, J. H. (1998). Location and the multinational enterprise: A neglected factor? *Journal of International Business Studies*, 29(1), 45–66.
- Earley, P. C., & Gibson, C. B. (2002). *Multinational work teams: A new perspective*. Mahwah, NJ: Lawrence Erlbaum Associates.
- Espinosa, J. A., Cummings, J. N., Wilson, J. M., & Pearce, B. M. (2003). Team boundary issues across multiple global firms. *Journal of Management Information Systems*, 19(4), 157–190.
- Foss, N. J., & Pedersen, T. (2002). Transferring knowledge in MNCs: The role of sources of subsidiary knowledge and organizational context. *Journal of International Management*, 8(1), 49–67.
- Foss, N. J., & Pedersen, T. (2004). Organizing knowledge processes in the multinational corporation: An introduction. *Journal of International Business Studies*, 35(5), 340–349.
- Foss, K., & Rodgers, W. (2011). Enhancing information usefulness by line managers' involvement in cross-unit activities. *Organization Studies*, 32(5), 683–703.
- Foss, N. J., Husted, K., & Michailova, S. (2010). Governing knowledge sharing in organizations: Levels of analysis, governance mechanisms, and research directions. *Journal of Management Studies*, 47(3), 455–482.
- Foss, N. J., Pedersen, T., Reinholt Fosgaard, M., & Stea, D. (2015). Why complementary HRM practices impact performance: The case of rewards, job design, and work climate in a knowledge-sharing context. *Human Resource Management*, 54(6), 955–976.
- Freeman, L. C. (1979). Centrality in social networks: Conceptual clarification. *Social Networks*, 1(3), 215–239.

- Freeman, L. C., Romney, A. K., & Freeman, S. C. (1987). Cognitive structure and informant accuracy. *American Anthropologist*, 89(2), 310–325.
- Gupta, A. K., & Govindarajan, V. (2000). Knowledge flows within multinational corporations. *Strategic Management Journal*, 21(4), 473–496.
- Haas, M. R., & Cummings, J. N. (2014). Barriers to knowledge seeking within MNC teams: Which differences matter most? *Journal of International Business Studies*, 46(1), 36–62.
- Hargadon, A., & Sutton, R. I. (1997). Technology brokering and innovation in a product development firm. *Administrative Science Quarterly*, 42(4), 716–749.
- Hennessey, B. A., & Amabile, T. M. (2010). Creativity. *Annual Review of Psychology*, 61, 569–598.
- Hinds, P. J., & Mortensen, M. (2005). Understanding conflict in geographically distributed teams: The moderating effects of shared identity, shared context, and spontaneous communication. *Organization Science*, 16(3), 290–307.
- Howard, G. S. (1994). Why do people say nasty things about self-reports? *Journal of Organizational Behavior*, 15(5), 399–404.
- Janssen, O. (2000). Job demands, perceptions of effort-reward fairness and innovative work behaviour. *Journal of Occupational Psychology*, 73(3), 287–302.
- Kellogg, K. C. (2014). Brokerage professions and implementing reform in an age of experts. *American Sociological Review*, 79(5), 912–941.
- Kilduff, M. J. (2004). The friendship network as a decision-making resource: Dispositional moderators of social influences on organizational choice. *Journal of Personality and Social Psychology*, 62(1), 168–180.
- Klueter, T., & Monteiro, F. (2017). How does performance feedback affect boundary spanning in multinational corporations? Insights from technology scouts. *Journal of Management Studies*, 54(4), 483–510.
- Kogut, B., & Zander, U. (1993). Knowledge of the firm and the evolutionary theory of the multinational corporation. *Journal of International Business Studies*, 24(4), 625–645.
- Krackhardt, D., & Hanson, J. R. (1993). Informal networks: The company behind the charts. *Harvard Business Review*, 71(4), 104–111.
- Kwon, S. W., & Adler, P. S. (2014). Social capital: Maturation of a field of research. *The Academy of Management Review*, 39(4), 412–422.
- March, J. G., & Simon, H. A. (1958). *Organizations*. New York: Wiley.
- McEvily, B., Soda, G., & Tortoriello, M. (2014). More formally: Rediscovering the missing link between formal organization and informal social structure. *The Academy of Management Annals*, 8(1), 299–345.
- McPherson, M., Smith-Lovin, L., & Cook, J. M. (2001). Birds of a feather: Homophily in social networks. *Annual Review of Sociology*, 27, 415–444.
- Mehra, A., Kilduff, M. J., & Brass, D. J. (2001). The social networks of high and low self-monitors: Implications for workplace performance. *Administrative Science Quarterly*, 46(1), 121–146.
- Minbaeva, D., & Santangelo, G. D. (2017). Boundary spanners and intra-MNC knowledge sharing: The roles of controlled motivation and immediate organizational context. *Global Strategy Journal*, 8(2), 220–241.
- Minbaeva, D., Pedersen, T., Björkman, I., Fey, C. F., & Park, H. J. (2003). MNC Knowledge transfer, subsidiary absorptive capacity, and HRM. *Journal of International Business Studies*, 34(6), 586–599.
- Monteiro, F., & Birkinshaw, J. (2017). The external knowledge sourcing process in multinational corporations. *Strategic Management Journal*, 38(2), 342–362.
- Monteiro, L. F., Arvidsson, N., & Birkinshaw, J. (2008). Knowledge flows within multinational corporations: Explaining subsidiary isolation and its performance implications. *Organization Science*, 19(1), 90–107.
- Mudambi, R., & Swift, T. (2009). Professional guilds, tension and knowledge management. *Research Policy*, 38(5), 736–745.
- Nebus, J. (2006). Building collegial information networks: A theory of advice network generation. *The Academy of Management Review*, 31(3), 615–637.
- Obstfeld, D. (2005). Social networks, the tertius iungens orientation, and involvement in innovation. *Administrative Science Quarterly*, 50(1), 100–130.
- Obstfeld, D., Borgatti, S. P., & Davis, J. (2014). *Brokerage as a process: Decoupling third party action from social network structure. Contemporary perspectives on organizational social networks*, Vol. 40, Emerald Group Publishing Limited 135–159.
- Ocasio, W. (1997). Towards an attention-based view of the firm. *Strategic Management Journal*, 18(5), 187–206.
- Podolny, J. M., & Baron, J. N. (1997). Resources and relationships: Social networks and mobility in the workplace. *American Sociological Review*, 62(5), 673–693.
- Podsakoff, P. M., & Organ, D. W. (1986). Self-reports in organizational research: Problems and prospects. *Journal of Management Studies*, 12(4), 531–544.
- Podsakoff, P. M., MacKenzie, S. B., & Podsakoff, N. P. (2012). Sources of method bias in social science research and recommendations on how to control it. *Annual Review of Psychology*, 63(1), 539–569.
- Podsakoff, P. M., MacKenzie, S. B., Lee, J.-Y., & Podsakoff, N. P. (2003). Common method biases in behavioral research: A critical review of the literature and recommended remedies. *The Journal of Applied Psychology*, 88(5), 879–903.
- Puranam, P., Raveendran, M., & Knudsen, T. (2012). Organization design: The epistemic interdependence perspective. *The Academy of Management Review*, 37(3), 419–440.
- Reagans, R., & McEvily, B. (2003). Network structure and knowledge transfer: The effects of cohesion and range. *Administrative Science Quarterly*, 48(2), 240–267.
- Reagans, R., Zuckerman, E., & McEvily, B. (2004). How to make the team: Social networks vs. demography as criteria for designing effective teams. *Administrative Science Quarterly*, 49(1), 101–133.
- Reagans, R., & Zuckerman, E. W. (2001). Networks, diversity, and productivity: The social capital of corporate R&D teams. *Organization Science*, 12(4), 502–517.
- Reinholt, M., Pedersen, T., & Foss, N. J. (2011). Why a central network position isn't enough: The role of motivation and ability for knowledge sharing in employee networks. *The Academy of Management Journal*, 54(6), 1277–1297.
- Roberts, M. J. D., & Beamish, P. W. (2017). The scaffolding activities of international returnee executives: A learning based perspective of global boundary spanning. *Journal of Management Studies*, 54(4), 511–539.
- Rogelberg, S. G., & Stanton, J. M. (2007). Introduction: Understanding and dealing with organizational survey nonresponse. *Organizational Research Methods*, 10(2), 195–209.
- Schotter, A., & Beamish, P. W. (2011). Performance effects of MNC headquarters–Subsidiary conflict and the role of boundary spanners: The case of headquarter initiative rejection. *Journal of International Management*, 17(3), 243–259.
- Schotter, A. P. J., Mudambi, R., Doz, Y. L., & Gaur, A. (2017). Boundary spanning in global organizations. *Journal of Management Studies*, 54(4), 403–421.
- Shalley, C. E., Gilson, L. L., & Blum, T. C. (2009). Interactive effects of growth need strength, work context, and job complexity on self-reported creative performance. *The Academy of Management Journal*, 52(3), 489–505.
- Shipilov, A. V., & Li, S. X. (2012). The missing link: The effect of customers on the formation of relationships among producers in the multiplex triads. *Organization Science*, 23(2), 472–491.
- Shipilov, A., Gulati, R., Kilduff, M., Li, S., & Tsai, W. (2014). Relational pluralism within and between organizations. *The Academy of Management Journal*, 57(2), 449–459.
- Soda, G., Stea, D., & Pedersen, T. (2019). Network structure, collaborative context, and individual creativity. *Journal of Management*, 45(4), 1739–1765. <https://doi.org/10.1177/0149206317724509>.
- Soda, G., Tortoriello, M., & Iorio, A. (2018). Harvesting value from brokerage: Individual strategic orientation, structural holes, and performance. *The Academy of Management Journal*, 61(3), 896–918.
- Soda, G., & Zaheer, A. (2012). A network perspective on organizational architecture: Performance effects of the interplay of formal and informal organization. *Strategic Management Journal*, 33(6), 751–771.
- Spiro, E. S., Acton, R. M., & Butts, C. T. (2013). Extended structures of mediation: Re-examining brokerage in dynamic networks. *Social Networks*, 35(1), 130–143.
- Stea, D., & Pedersen, T. (2017). Not all brokers are alike: Creative implications of brokering networks in different work functions. *Human Relations*, 70(6), 668–693.
- Stea, D., Pedersen, T., & Foss, N. J. (2017). The relational antecedents of interpersonal helping: 'quantity', 'quality' or both? *British Journal of Management*, 28(2), 197–212.
- Thompson, J. D. (1967). *Organisations in action*. New York: McGraw-Hill.
- Tortoriello, M., & Krackhardt, D. (2010). Activating cross-boundary knowledge: The role of simmelian ties in the generation of innovations. *The Academy of Management Journal*, 53(1), 167–181.
- Tortoriello, M., Reagans, R., & McEvily, B. (2012). Bridging the knowledge gap: The influence of strong ties, network cohesion, and network range on the transfer of knowledge between organizational units. *Organization Science*, 23(4), 1024–1039.
- Tsai, W. (2001). Knowledge transfer in intraorganizational networks: Effects of network position and absorptive capacity on business unit innovation and performance. *The Academy of Management Journal*, 44(5), 996–1004.
- Tushman, M. L., & Scanlan, T. J. (1981). Boundary spanning individuals: Their role in information transfer and their antecedents. *The Academy of Management Journal*, 24(2), 289–305.
- Wasserman, S., & Faust, K. (1994). *Social network analysis*, Vol. 8. Cambridge: Cambridge University Press.
- Xiao, Z., & Tsui, A. S. (2007). When brokers may not work: The cultural contingency of social capital in Chinese high-tech firms. *Administrative Science Quarterly*, 52(1), 1–31.
- Zenger, T. R., & Lawrence, B. S. (1989). Organizational demography: The differential effects of age and tenure distributions on technical communication. *The Academy of Management Journal*, 32(2), 353–376.
- Zhou, J., Shin, S. J., Brass, D. J., Choi, J., & Zhang, Z.-X. (2009). Social networks, personal values, and creativity: Evidence for curvilinear and interaction effects. *The Journal of Applied Psychology*, 94(6), 1544–1552.